

Comparative Analysis of Ensemble and Neural Network Models for Food Waste Prediction

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Abstract— Food waste is a major problem in the United States. It can lead to economic losses and negative environmental impacts. Predicting food waste helps policymakers and businesses plan better and reduce waste. This study used data from the ReFED Food Waste Monitor. After preprocessing, the dataset contains 75,101 records and 13 variables, focusing on year, state, sector, sub-sector, food type, surplus and supply amounts, uneaten food, environmental footprints, meals wasted and total food waste as the target variable. We applied four predictive models including XGBoost, Random Forest, Feedforward Neural Network (FNN) and Knowledge-Informed Neural Network (KiNN). The models were compared using R^2 , RMSE and MAE to measure prediction accuracy. Results showed that Random Forest performed the best, achieving the highest R^2 (0.9996) and the lowest RMSE (10.62) and MAE (1.16), indicating predictions that closely match the actual food waste values. XGBoost ranked second, and FNN performed better than KiNN. These results highlight the main factors that influence food waste and show that ensemble models can provide more reliable predictions than neural networks. The findings can support sustainable decision-making by helping identify priority areas for waste reduction and guide policies to reduce food loss across sectors and regions.

Keywords— Food waste prediction; Sustainability analytics; Ensemble models; Neural networks; Decision support systems

I. INTRODUCTION

Food waste is defined as food that is produced for human consumption but is discarded or unconsumed across the food supply chain. This problem has become a major challenge especially for sustainability because of its impact on not only the environment, but also the economy, and society as it causes wasted resources and financial losses [1]. This is all happening alongside widespread food insecurity. However, wastage of food generation is very inconsistent and varies by region, sector and food type. Thus, making it hard to manage with general policy measures.

Food waste is a very serious issue especially in the United States. Over 59.8 million tons of food are being wasted and generating over 70.7 million tons of surplus food [2]. This leads to significant economic losses and environmental harm from landfill emissions. These emphasize the need for data-driven analysis to create plans that are targeted towards sustainability and strategies to reduce food waste.

A. Research Problem

Food waste production in the United States highlights great differences between the states, economic sectors, and food categories. Thus, leading to differences in production methods, consumption habits, and waste management systems. This variation limits the general policy measures and also sustainability planning becomes more complicated. Therefore, the research problem of this study is a need for a reliable, data-driven predictive model that can predict food waste that is based on different states, economic sectors, and food types. A solution to this problem will support better decision-making and focused waste reduction strategies.

B. Research Objectives

The objectives of this research are to predict the total amount of surplus food that becomes waste in the United States by considering factors such as the location where the waste occurs, the main food system segment, the detailed category within each segment, the specific type of food and quantitative measures of food production, supply and consumption, as well as environmental impacts. The study aims to build and compare the performance of predictive models, including ensemble models and neural networks to determine which approach performs the best in predicting food waste. Additionally, the research aims to identify the variables that have the greatest influence on food waste in ensemble models and to develop a forecasting tool that can support sustainability planning and strategies for reducing food waste at both regional and national levels.

C. Research Questions

1. How accurately can the total amount of surplus food that becomes waste be predicted using factors such as state, sector, sub-sector, food type, surplus amounts, supply, uneaten food and environmental footprint variables?
2. How do ensemble models compare with neural network models in predicting food waste levels?
3. Which factors or variables contribute most to food waste generation across U.S. states and food system sectors?
4. How can a prediction or forecasting tool support sustainability planning and help policymakers design effective waste-reduction strategies?

D. How data mining enables sustainable decision support

Data mining is helpful in the discovery of patterns and generating insights from large and distributed data. Data

related to food waste issue usually consists of thousands of rows and many different attributes, so it is not realistic to analyse without using any computational methods. Data mining is needed to help identify patterns, determine key factors contributing to food waste, and predict future waste levels for future prevention [3]. The insights generated can be used for policymakers to suggest ways to minimize wastes and carbon emissions so that environmental sustainability can be improved. Social sustainability can also be achieved by giving the food surplus to the people or communities in need of these foods. For economic sustainability, businesses and organizations can save costs as they are able to plan their inventory based on the food demand and supply from the insights obtained. This can prevent overproduction and the costs saved can be used in other areas that need advancement [4]. Overall, data-driven models transform large volumes of raw data into meaningful insights, allowing stakeholders to make informed decisions that address current needs at the same time supporting long-term sustainability.

E. Overview of Paper Structure

This paper has been organised to follow the subsequent structure to present the study in a systematic and methodical way. The first section will delve into the introduction, where the research problem and questions were highlighted as well as the functionality and sustainability of data mining into supporting decision making. Next, the second section involves literature review in which journal articles regarding topics of food waste, machine learning models and data mining techniques were researched and examined. The third section, research methodology, explored the stages of data mining including problem and data understanding, data pre-processing, modelling techniques and evaluation strategy. The fourth section discussed the results of the various models developed which are XGBoost, Random Forest, Feedforward Neural Network (FNN) and Knowledge-Informed Neural Network (KiNN), via the evaluation strategy. These models are then compared and evaluated to extract findings as well as possible limitations and issues encountered. Finally, the fifth and final section summarized the findings and how they would contribute to solving the food waste issue through better decision making. This section also acknowledged possible limitations within the study, offering recommendations for improvements in future studies.

II. LITERATURE REVIEW

Solutions that are sustainable and scalable to solve the issue of food waste which is a significant economic and environmental problem is needed. Previous research has used machine learning and neural network models to study patterns for food waste and enhance predictive accuracy. This section is a review of the literature that considers current approaches, their limitations and research gaps, which support the choices made in this study.

A. Food waste

This study examines the growing issue of perishable food waste in retail supermarkets [5]. They mostly focus on fresh

food categories such as vegetables, bakery items, meat, fish, and fruit in New Zealand. The studies highlight that supermarkets in New Zealand produce about 60, 500 tonnes of food waste each year. Vegetables and bakery products take a huge portion of the food waste. Few main reasons why waste occurs is due to demand forecasting and overstocking, which can reach up to 30%. Ensemble and neural networks helped to reduce food waste by 14% to 52% and improved revenue and efficiency of operations. This study concludes that data-driven decision making and machine learning together can offer a strong strategy to not only reduce food waste but also support sustainability goals and enhance retail planning and resources use.

B. Machine Learning and Ensemble Models for Food Waste Predictions

This study focuses on predicting food wastage level using machine learning techniques [6]. This study used a comparative approach to predictive modelling by testing five machine learning algorithms. The algorithms used are AdaBoost, XGBoost, CatBoost, Support Vector Machine with Gaussian Process, and Elastic Net Regression. A dataset that includes variables such as type of food, quantity prepared, seasonality, condition of storage, pricing, and event-related factors was used for analysis. The results showed that XGBoost outperformed the other algorithms by achieving the lowest prediction errors with the values of $MSE = 31.05$, $RMSE = 5.5.7$, $MAE = 4.13$. It also achieved the highest explanatory power with the R-square value of 0.73. This results highlights the strong relationship between the predicted and actual food waste values. Factors like guest count, seasonality, location, and food type significantly influence food waste. The limitation of this study is that the model prediction performance is sensitive to the availability and quality of data, and as such, the model may have limited robustness to variables across not similar food waste contexts. Overall, the study concludes XGBoost as a strong and dependable model for predicting food waste.

Ifham et al. noticed the issue of food waste in Sri Lanka due to the lack of awareness in society and poor planning that causes overproduction and eventually contributes to spoilage and wastage [7]. Random Forest Regressor, Decision Tree Regressor and Gradient Boosting Regressor were used to predict food wastage. The data partition for training and testing sets was divided in a ratio of 80:20. MSE, RMSE and R^2 were used to evaluate the performance of the models. Results showed that Random Forest Regressor is the most effective model in predicting food wastage. It has the lowest MSE value of 7.41 and RMSE value of 2.72, with the highest R^2 value of 0.929. A key strength of the study is that it covers multiple aspects of food waste, including predicting demand and ingredient requirements, near-expiry food prices, and optimal recycling methods for expired food. It uses different machine learning algorithms to predict different outcomes, which makes the performance reliable. However, broader dataset coverage could be included, like more data from different restaurants, catering services and households across various regions could improve generalizability.

Rodrigues et al. used machine learning algorithms to predict food demand to minimize food waste and avoid underproduction [8]. Random forest, gradient boosting machine algorithms, long short-term memory (LSTM) neural network and transformer neural network were used as comparison. The algorithms are tested on different dataset from three food catering services (two student canteens and a company canteen) to see how different algorithms affect the quality of prediction. Nested cross-validation are applied in every model. Models are evaluated using RMSE, MAPE and R^2 and the results are compared against the baseline models. Results showed that random forest models perform the best for student canteens datasets, LSTM neural network performs the best for company canteen datasets. This shows that different food services settings require different forecasting models. A key strength of the study is its recognition of contextual differences because different models performed best in different food service settings. However, the findings are limited in generalizability since the analysis was based on only three food catering services.

Turker stated that the campus lacks proper food service management, leading to significant food waste, which in turn increases operational costs and contributes to environmental degradation [9]. In this paper, the machine learning algorithms used are Gradient Boosting Regressor, XGBoost Regressor, Linear Regression, Lasso Regression, Decision Tree Regressor and Extra Tree Regressor. Datasets were obtained from open-source platforms and include information such as food types in the dining hall, campus mobility, number of meals, daily activities, and more. Feature extraction and standardization were performed to improve data quality. Cross-validation was applied, and the models' performance was evaluated using R^2 , MAE and RMSE. Based on the results, Lasso Regression achieved the highest R^2 value of 0.999, whereas XGBoost Regressor came second with an R^2 value of 0.882. The Lasso Regression model reduces food waste by 28% and improves operational efficiency, menu planning and resource allocation. The study is lacking long-term data which constrains the model's ability to generalize across different time periods, seasonal variations and unusual events.

Yang proposed that machine learning can solve the issue of food waste across different fields through prediction using Random Forest and indicators such as food industry, commercial demand, household behaviour [3]. The results found that household behaviour is highly correlative to the target variable with a value of 0.731 while commercial and food industry estimates showed medium association. This indicated that food waste coming from households has the most influence towards food waste precision, which also indicate that household contributed a lot in the food waste issue compared to the commercial or food industry. In terms of the model performance, the Random Forest performed well showing 95% accuracy score and 0.97 precision score, indicating the model's ability to accurately determining positive causes from the false positive cases. Moreover, the model was also able to identify 93% of actual positives and had a 0.95 F1 value, revealing that the model identifies true positives while reducing false positives.

Seyam et al. designed a demand forecasting framework and created using an ensemble stacking model [10]. This model integrated base learner models which are eXtreme gradient boosting, support vector regression, random forest, long short-term memory models and used Ridge regression for the meta-learner. The findings of this research showed that the stacking ensemble model accomplished a high accuracy score of 94.44%, 0.63 MAE, 1.8 for MAPE and R square value of 0.99, performing slightly better than the random forest and eXtreme gradient boosting. Support vector regression and long short-term memory model performance are also repeatedly surpassed by the ensemble model. Additionally, XGBoost excels when it comes to efficiency in computing capabilities while Stacking would be the better option in terms of accuracy. A possible limitation highlighted was that a small subset of synthetic dataset was used to test the model and a lack of weather-related variables. Suggestions for future studies stated that real-life and large datasets should be used to test stacking models' performance and impact to this food waste issue.

Shopnil et al. aimed to predict food wastage in students' hostels by applying machine learning models to study the factors that affect food waste and propose the best ensemble model [11]. The study developed an ensemble machine learning-based food waste prediction system using Random Forest, Logistic Regression, Support Vector Machine, Decision Tree, Gaussian Naïve Bayes and AdaBoost. All possible combinations of these models were tested to identify the best ensemble model. The results showed that Random Forest and Decision Tree achieved the best F1-scores among individual models, while the ensemble model combining Random Forest and AdaBoost achieved the highest overall F1-score of 86.88%. Predictive modelling can effectively forecast food waste and support sustainability planning. The article successfully uses ensemble models on real-world data to achieve high prediction performance and provide useful and accurate insight in improving food waste management and sustainability. However, the model focuses on classifying categories of food waste rather than predicting the actual amount of waste in continuous values such as kilograms or tons. Based on the gap, our research predicts actual food waste amounts to better support sustainability planning decisions.

C. Neural Network Approaches for Food Waste and Food Loss Prediction

This study by Lubura et al. explores food recognition and food waste prediction using convolutional neural networks (CNNs) to create an automated and accurate way to quantify food waste from meal images [12]. This study shows that a deep learning-based computer vision framework was developed to train a CNN to recognize 157 food categories. Later, a comparative analysis was performed of plates before and after meals to estimate waste percentages. The CNN shows great results by achieving a test accuracy of 98.8% and a loss value of 0.102 after training reflecting its strong food recognition ability. When applied to real-world data collected from Serbian students, the study predicted an average food waste of 21.3% per meal that underscores the significant amount of avoidable waste. However, the accuracy of the

proposed CNN-based food waste estimation is easily affected by the varying conditions in an image including plate shape, color, background clutter and existence of non-food objects, making it challenge to be applicable to real dining environments. However, it will also help acknowledge practical limitations related to plate shape, background interference, and non-food items.

Sun et al. proposed an intelligent food waste identification and analysis system via adjusting the convolution neural network (CNN) and improved using fine-tuning techniques through the application of new training dataset and output layer in the neural network. [13]. The study also implemented four Internet of Things (IoT) layers including the sensing layer, network layer, data storage layer and application layer for higher efficiency in collecting and storing data. The study compared the results of the initial and improved CNN model, revealing an 82.3% decrease in time to train the model and 96.3% reduction in parameters required to be trained as well as an increase in accuracy from 94.7% to 97.6%. Moreover, the findings also revealed that the accuracy of the CNN model that applied fine-tuning methods increased to 94.78% which is a strong rise compared to the initial accuracy of 36.16% without fine-tuning. Thus, the study concluded that the use of fine-tuning techniques significantly increases the accuracy, training time efficiency and decrease the number of trained parameters of the CNN model. Recommendations for future studies were also highlighted, stating the need to explore attention mechanisms to reduce cost of resources and improve CNN model performance.

Machine learning can help predict food loss and waste based on socio-economic and behavioural factors [14]. They applied two machine learning models using statistical data collected from surveys in Romania. The decision tree built a tree structure by splitting data based on features. The neural network contains hidden layers that used ReLU activation function to capture nonlinear patterns among all features and output layer that used Softmax activation to classify households into food waste categories. Results showed that the neural network model outperformed the decision tree in both accuracy (78%) and robustness with higher MCC value (0.70) that measures the binary classification quality. The study explained that the decision tree is simpler but unlike the neural network that can effectively capture complex patterns and interactions. The study also highlighted that household socio-economic factors strongly affect food waste and future work should quantify their exact impact. The study's strengths include using advanced models and comparing different methods to predict food waste on consumer behaviour, while its limitation is they may not consider other important factors such as environmental. Our research can address this gap by considering more factors such as supply chain and environmental factors.

Guo et al. aimed to predict postharvest food loss at national and sub-national levels using neural network models [15]. FNN and KiNN are applied with FAO Food Balance Sheets and World Bank data such as year, population, food production, GDP per capita, precipitation, protein supply and production of the referred crop. FNN learns the relationship between inputs and postharvest food loss by adjusting its

weights to minimize prediction error during training. KiNN extends this approach by using expert knowledge into the learning process and follows simple food system rules to guide training. The results show that FNN performs better with the full dataset while the KiNN outperforms the FNN when both are trained on a smaller subset. This indicates that FNN is more suitable for large datasets, whereas KiNN is more effective for limited data. They also found that removing Year, Precipitation, GDP or Production caused large drops in accuracy. In contrast, removing FAO Region or Population had little effect, meaning that these were either redundant or not relevant while removing Protein Supply slightly reduced performance. The study highlights the strengths of FNN and KiNN but model accuracy may be lower for small areas or real-life settings because FAO data are country-level averages rather than direct measurements.

III. RESEARCH METHODOLOGY

This research follows a systematic end-to-end data-driven approach predicting levels of food waste to assist sustainable-based decision-making. The analytical mechanisms combine systematic preprocessing of data, robust learning from machines, and modelling with neural networks, as well as rigorous model validation to guarantee the reliability and interpretability of results. A flow diagram is used to illustrate the research pipeline from raw data retrieval to model evaluation and result interpretation. Both the data preprocessing and model training was done using the Google Colab platform and the code is available in GitHub where the link to it is under Appendix C.

A. Process Flow

Figure 1 illustrates the flow of the research conducted in this study which can be seen under Appendix A.

B. Problem and Understanding

To predict food waste amounts, the dataset is collected from the ReFED Food Waste Monitor, a national database that tracks food production, consumption, waste, donations and losses across the U.S. [2]. The raw dataset can be referred to Appendix C. It contains 88,281 of data with a total of 31 variables. The description of the 31 variables is shown in Table 1. This dataset is suitable for predicting food waste amounts for sustainability planning because it provides detailed, reliable and high-quality information covering multiple aspects of food production, consumption and waste. The dataset is complete and well-structured that allows us to build predictive models such as ensemble models and neural network models. This helps policymakers predict food waste, plan resources efficiently, reduce waste and mitigate environmental impacts. From a decision-support perspective, this dataset helps policymakers and stakeholders identify areas with high food waste, compare waste levels across different states and sectors and predict future food waste amounts. This information can support better decisions on how to allocate resources, support waste-reduction strategies and plan sustainability policies to reduce food waste and its environmental impacts.

C. Data Pre-processing

The data preprocessing was begun by downloading the raw food waste dataset from the ReFED. However, it could not be directly used as it is to train the models due to problems with the headers. Hence, smart header detection was employed to find the proper column keys.

Feature selection retained variables directly correlated with food waste generation, and sustainability decision making. Numerical variables, which are `tons_surplus`, `tons_un eaten`, and `tons_waste` was selected to measure waste magnitude. Adding environmental indicators such as `surplus_total_year_mtCO2e_footprint`, `surplus_total_100_year_mtCH4_footprint` and `gallons_water_footprint` helped identify relationships between amounts of waste and sustainability impacts. The categorical features such as `state`, `sector`, `sub_sector` and `food_type` offer context of geographical location mismatch in operational profiles and type of product to explain the varying food waste response. The variables that were not directly related to the generation of food waste or sustainability impact have been removed to reduce noise and increase the efficiency of the model which ultimately allows further interpretation. Additional variables were included as supporting indicators of food availability and consumption-scale effects using the variables `tons_supply` and `meals_wasted` as it indirectly influences food waste generation.

All numeric columns were forced to numeric data types with NaN substituted for non-numeric to ensure consistency. Missing values were then removed to preserve the integrity of the dataset. Rows with zero values were also removed as they were considered invalid for prediction models.

The model was trained after the dataset was trimmed from 88, 281 rows to 75, 101 rows after removing the extreme outliers by applying the 1st – 99th percentile clipping. While this makes the model more stable, it may exclude some extreme but valid examples. Furthermore, the model was trained with log-transformed target values to stabilize variance and achieve robustness against high deviations. Cleaned dataset was uploaded in GitHub under Appendix C.

D. Modelling Approaches

The algorithms used to predict the food waste level include two ensemble models, which are XGBoost and Random Forest, and two neural network models, which are Feedforward Neural Network (FNN) and Knowledge-Informed Neural Network (KiNN).

XGBoost is an ensemble model that can capture complex non-linear relationships in features like food type and sector. It performs well in handling noisy real-world data which is common in food waste datasets. In this study, the model builds 500 trees with a learning rate of 0.05. Each tree can grow up to a maximum depth of 9. The model uses all rows and 80% of the features when building each tree. These settings were manually tuned to balance accuracy and overfitting.

Random Forest is an ensemble model that can combine results from many decision trees to capture complex patterns. In this study, the model uses 200 trees with no maximum depth, so trees grow fully until leaves are pure. It uses bootstrapping, default minimum samples per split and leaf, and automatically selects features for each split. Parallel processing is enabled to speed up training.

FNN is a neural network in which data flows in one direction. The hidden layers help to learn non-linear relationships between multiple input factors and waste levels. In this study, the FNN contains two hidden layers with 128 and 64 neurons, using the ReLU activation function. It trains with the Adam optimizer, learning rate 0.001, batch size 256, and up to 100 epochs. To prevent overfitting, it uses a dropout rate of 0.2 and early stopping after 5 epochs. Performance is checked with 5-fold cross-validation.

KiNN is a feedforward neural network augmented by domain knowledge in its design. It includes categorical domain information with embedding layers. These learned embeddings encode some relationship and similarity between categories which is not included in the raw format. KiNN is suitable for food waste modelling, where contextual and categorical factors play a critical role in determining waste levels. In this paper, KiNN has three hidden layers with 128 and 64 neurons, plus four embedding layers for categorical features. It uses ReLU activation, the Adam optimizer with 0.001 learning rate, and a dropout rate of 0.2. The target is log-transformed, and numerical features are standard-scaled. Training stops early if validation loss doesn't improve for 5 epochs, and performance is validated with 5-fold cross-validation.

All four models help predict the total tons of food waste, achieving objective 1, and allow a comparison between ensemble and neural network approaches to analyse effectiveness, achieving objective 2. They also help achieve objective 3 by highlighting which factors contribute most to food waste. Finally, the resulting model can form the basis of a forecasting tool to guide resource allocation and waste-reduction policies, achieving objective 4.

E. Evaluation Strategy

To evaluate the four ensemble and neural models, performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Adjusted R squared (R^2) were selected. The metrics are commonly used in regression evaluation for scikit-learn.

MAE was chosen due to its interpretability and is less sensitive to outliers. It measures the mean of the absolute discrepancies between predicted and actual values by calculating residual average. RMSE was also used as it helps identify extreme cases of food waste by penalising extreme values. RMSE evaluates how accurate the regression or machine learning model is by calculating the alignment between the predicted values and actual values of the dataset. R-squared, also known as coefficient of determination, is commonly used in statistics to measure the goodness of fit and explanatory power of models. The metric was useful in

evaluating our models as it helps measure the overall model fit and allows comparison between the ensemble and neural networks by being scale independent.

Besides using metrics, 5-fold cross-validation was also used to equally partition the dataset into 5 subsets. This helps introduce less bias as the dataset will be divided into 1 validation subset and 4 training subsets. The process was repeated five times ensuring that each subset will be used for validation once. Results of all 5 folds will then be averaged to find the final value.

Model-specific input transformations were also implemented for the ensemble models and neural models. This helps ensure the dataset is appropriate for the learning characteristics of each technique. For the neural network models, feature scaling was implemented to prevent extreme features from causing imbalances during training. Logarithmic transformation for the target variable was also required for the neural networks to help reduce the skewness and balance any difference in the food waste values. Meanwhile for the ensemble models, one-hot encoding for the categorical variables were included to treat each category independently, allowing trees to split on category presence. Evaluation of model performance is a crucial stage in machine learning development to gauge the accuracy and reliability of the model.

IV. RESULTS AND DISCUSSION

A. Model Performance

1) XGBoost Model

Based on Table 2, the model has R^2 of 0.9996 which is very close 1. This means that the model explains almost all the variability in tons_waste and captures nearly all the patterns in the data. RMSE of 10.7377 shows that on average, the model's predictions are about 10.74 tons higher or lower than the actual food waste amount. RMSE is sensitive to large errors, so this higher value indicates that a few predictions are much farther from the true values. The MAE is 2.375, which means that on average, the model's predictions are off by only 2.38 tons. This shows that most predictions are very close to the actual values, with only a few larger deviations leading to the higher RMSE.

Table 3 and Figure 2 show the top ten most important features in predicting food waste in tons using the XGBoost model. meals_wasted has the highest importance scores among all variables, which shows that the amount of food lost strongly reflects how much food is eventually wasted. tons_un eaten is the second highest, which shows that uneaten food also strongly affects the food waste amount. The retail sector rankstons_un eaten is the third highest, which shows that uneaten food also strongly affects the food waste amount. The retail sector ranks third, which shows that waste generated at the retail stage contributes more compared to other sectors such as manufacturing and residential. Figure 3 shows the five features with lowest importance. This shows that military,

recreation, healthcare and lodging sub-sectors, as well as the state of Ohio, have a very little impact on model's prediction.

Overall, the XGBoost model performs very well in predicting the food waste amount. It explains almost all the variation in the data and most predictions being close to the actual values. The model also clearly identifies the most important factors influencing food waste, especially food that is wasted and left uneaten. It is effective and reliable for supporting food waste analysis and sustainability planning.

2) Random Forest Model

The random forest model showed strong performance across all metrics as shown in table 4. Firstly, the average MAE has a value 1.16 tons showing high precision as the model only showed wrong predictions of about 1.16 tons of food waste. Given that food waste can differ between different sectors and categories, the result showed relatively good predictive outcomes. Meanwhile, the RMSE of 10.62 showed the cases where prediction is off. R^2 of 0.9996 shows how the model was able to explain 99.96 percent of the changes in the food waste.

Meanwhile, feature importance was also conducted. Table 5 shows the result, indicating that meals_waste and tons_un eaten had the highest influence as predictors for food waste as they account for 51.5 percent and 46.0 percent of the total importance. The result also showed that sector (cat_sub_sector_Not Applicable_ with value of 0.000063) and food type (cat_food_type_Breads & Bakery with value of 0.000063) had lowest importance. This suggests that sectors and types of food had low explanatory power for food waste prediction. Bakery products tend to have shorter shelf life, making these goods prone to spoilage compared to other products. Meanwhile, sub-sector provided limited predictive value due to lack of categorisation, thus records were not linked to a specific sub-sector. The overall results showed that the Random Forest model depends on direct consumption-related variables to predict the total food waste.

The random forest model was ideal for food waste modelling as it is an ensemble model that uses bagging to aggregate the predictions from multiple decision trees training on different subsets of data. The model reduces overfitting and improves generalisation. Random forest can capture non-linear relationships between different features while also providing feature importance calculations.

3) Feedforward Neural Network (FNN)

Table 6 shows the model performance of FNN. The model obtained a R^2 value of 0.9777. R^2 is used to show how close the model's prediction matches the real data. 0.9777 is very close 1 which means that the model capture almost all the patterns and relationships in the data. In other words, the FNN model shows its great ability in realistically estimate the amount of food waste for different products and areas.

The model also achieved a MAE value of 25.02. MAE shows how much on average the models can get wrong when

predicting food waste. Therefore, the value tells that FNN's predictions on average are about 25 tons higher or lower than the actual food waste. Furthermore, FNN obtained a RMSE value of 75.57. This indicates that for those situations where the model provides very wrong estimates, it can provide predictions that is about 76 tons difference from actual value.

In short, these metrics shows that FNN is effective in capturing the overall patterns of the food waste but the high MAE and RMSE show that the predictions can be off by a large amount and can create a big difference with the actual value.

4) *Knowledge-Informed Neural Network (KiNN)*

According to table 7, the KiNN model shows good predictive performance by having a R^2 of 0.9324, which means that more than 93% of the variability in food waste can be explained by the model. On average, the prediction of food waste of the KiNN model differs from what it actually predicts for being observed with the MAE value of 46.38 tons. This provides a simple, intuitively appealing measure of the extent to which they can predict and forecast income more accurately. The smaller MAE values indicate the closer to the prediction is to the real values. The RMSE was 133.45 tons, which is much larger compared to MAE. This indicates that although most predictions are close to the real values, there exist some cases where the model's errors are significantly larger. Larger deviations like these are common to occur because of the natural fluctuation in food waste data and limited quantity cases.

These statistics indicate that KiNN may be able to reliably predict common quantities of food waste, thereby informing planning and resource allocation, while also showing that in extreme cases additional consideration will be necessary.

In general, KiNN is a good model for food waste prediction according to the trade-offs between predictivity and interpretability. Its capability to utilize both structured numeric and categorical information is well-suited to data-driven sustainability interventions.

B. *Comparison of Models*

Table 8 presents a comparison of performance results across XGBoost, Random Forest, FNN and KiNN. Both XGBoost and Random Forest achieved high predictive accuracy with R^2 values of 0.9996. This means that they explain nearly all the variation in food waste and capture almost all patterns in the data. Their RMSE and MAE values were also low, which shows that most predictions closely matched the actual food waste amounts. Among these two ensemble models, Random Forest had slightly lower RMSE and MAE. This suggests that it has better overall prediction performance in this dataset compared to XGBoost. These results demonstrate that ensemble models are highly effective for predicting food waste. This may be due to their ability to capture complex interactions between numeric and categorical features such as location, sector, sub-sector and

environmental footprint variables. In contrast, the FNN and KiNN performed less accurately. FNN achieved an R^2 of 0.9777, with high RMSE (75.57) and MAE (25.02), while KiNN had the lowest performance. These larger errors show that neural network models are less effective than ensemble models in capturing complex patterns in the dataset. This may be due to the dataset size or the mix of categorical and numeric variables.

Overall, ensemble models are the most suitable choice, with Random Forest showing the best overall predictive performance in predicting food waste and supporting sustainability planning. These models provide actionable insights for waste-reduction strategies, whereas neural networks, particularly KiNN, are less reliable in this context.

C. *Interpretation Toward Sustainable Goal*

Random Forest has the best overall performance with a high R^2 value, and low RMSE and MAE. This makes it suitable to use as a tool to assist stakeholders when making decisions related to sustainable food waste management. Accurate predictions allow management to allocate resources in the most efficient way possible to decrease waste and reduce environmental impact. By predicting food waste, organisations can adjust their planning in food production and food waste management to avoid producing high amount of waste.

A feature importance analysis is done to find out which features have the most impact on the prediction. It shows that variables like `meals_wasted` and `tons_un eaten` are the most influential predictors. This indicates that consumption-related factors affect food waste more than other variables like sector or food type alone. Therefore, policymakers should focus their proposals on strategies like portion control, better demand forecasting, and improve inventory management instead of over-relying on sector-specific interventions.

Random Forest is effective in helping stakeholders to identify areas with a high risk of waste in advance. They can propose strategies to solve food waste issues beforehand such as redirecting extra food through donation programs or transforming them for other usage before becoming waste. By doing so, environmental burden can be reduced and social sustainability is improved.

In short, the Random Forest model can interpret complex data regarding food waste effectively. However, it is also necessary to apply knowledge from experts while using this model and complemented with other tools as well. This will help to achieve reliable results regarding extreme situations. By combining data-driven predictions with human judgment and coming up with adaptive strategies, stakeholders can help mitigate limitations and improve overall sustainability outcomes within the food system.

D. *Unexpected Findings*

Several surprising observations were made during analysis and interpretation of the data from this study. Although XGBoost and Random Forest statistics

demonstrate strong predictive accuracy, the predictive performance was already higher than initially anticipated based on historic complexity and variability of food waste data. Numeric variables directly connected to waste generation, such as `meals_wasted` and `tons_uneaten`, are strongly drive the prediction task while categorical variables such as `food_type` and `sub_sector` had less impact.

Next, fully optimized neural network models did not perform as well as ensemble models even after feature scaling and extensive model architecture tuning. MAE and RMSE scores of FNN and KiNN are considerably larger compared to XGBoost and Random Forest suggesting they had a more difficult time capturing large variations and outliers. Moreover, the performance of KiNN was not significantly better than a standard FNN due to the dataset is relatively large even after outliers are removed, reducing relative advantage on the embedding-based categorical representations.

Removing the extreme outliers by omitting the bottom 15% of data through 1st-99th percentile filtering greatly improved training stability and decreased variance for neural network models. This was consistent with the larger RMSE of FNN and KiNN, implying that for majority cases performance solidified while being degraded on extreme values. This raises a trade-off between model robustness and real-world representativeness for sustainability-oriented prediction tasks.

Overall, these unexpected findings reveal that simpler ensemble models can outperform more advanced neural architectures for structured food waste, offering crucial considerations for model selection in food waste forecasting and sustainability decision support.

E. Limitation and Challenges

The dataset used for this study was based in the US which may cause generalisation of the findings given that the model may not be applied to context from outside the country for training and evaluation. Food waste may differ in terms of location, culture, economics and even social habits. Additional food waste determining factors that differ in each country also includes supply chain, consumer patterns, pricing and management framework. This indicates that the ML models developed in this study may not be as effective or accurate for food waste prediction in geographic locations outside the US.

A possible constraint to this study is that both the ensemble models, XGBoost and Random Forest performed exceptionally high R^2 values, indicating high predictive performance which may suggest potential risk of overfitting or data leakage. Given that some of the variables are directly correlated to the target variable, this may suggest possible encoding of the outcome. Hence, the model performance may be biased and perform less effectively with datasets that do not have similar variables.

Despite the high-performance capability of the Neural Network models, FNN and KiNN, may not always be the most appropriate models to be applied when it comes to structured data. Given that neural networks are more suited for unstructured datasets such as images, text or audio, tabular data tends to consist of well-defined numerical and categorical variables. The results consistently indicate that ensemble-based models can outperform the neural network models across all performance metrics. This may be due to the ability of the ensemble tree-based models to identify non-linear relationships between variables without additional scaling or feature change. Hence, more complex models do not necessarily indicate improved model performance.

V. CONCLUSION

This study has developed and compared four predictive models, including XGBoost, Random Forest, FNN and KiNN to predict food waste. Results show that Random Forest has the best performance among all models. Meanwhile, the KiNN model is least effective in capturing large variations. From the feature importance analysis in Random Forest, it shows that the `meals_wasted` and `tons_uneaten` affect the model prediction the most. Thus, these two areas can be the key areas for intervention to greatly reduce food waste issues.

This study demonstrates the importance of accurate prediction of food waste in making long-term sustainable decisions and improvements. Through accurate prediction, organizations are able to manage their resource allocation, make appropriate planning based on the predicted food demand, and propose effective approaches to address food waste issues. By doing so, organizations can achieve continuous improvements in reducing food waste and supporting long-term sustainable practices.

There are a number of limitations in this study. First, the dataset contains only US-based data which minimizes the generalizability of the results. Second, the good performance of ensemble models could be a sign of overfitting due to the strongly correlated input variables with the target, the models may obtain lower accuracy when other datasets are used. Lastly, neural network models can learn complicated patterns, but they are less effective in handling structured tabular data. The also indicate that complicated models do not necessarily perform better.

For future work of this study, more predictive algorithms can be used as comparison to understand their effectiveness and performance in different food waste data. In addition, the research on how food waste is affected by factors in other areas can also be included. For example, how consumer behaviour or policy interventions can affect food waste. Moreover, hybrid models can be developed by combining two or more techniques to exploit their strengths and minimize their weaknesses. Lastly, the methodology may also be applied to other fields, including energy management or logistics systems, to expand its practical use and help in the planning of long-term sustainability across various areas.

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APPENDIX A

Table 1: Description of Variables in Dataset

No	Variable	Data Type	Description
1	year	Numeric	The year in which the food surplus and waste measurements were recorded.
2	state	Categorical	The U.S. state where the food surplus or waste occurred.
3	sector	Categorical	The broad segment of the food system where surplus or waste was generated.
4	sub_sector	Categorical	A more specific classification within the main sector.
5	sub_sector_category	Categorical	A further breakdown of sub_sector and group similar activities or products into one category.
6	food_type	Categorical	The specific type of food associated with the surplus or waste (e.g., frozen, produce, dry goods).
7	food_category	Categorical	A higher-level grouping of food types (e.g., eggs, cheese).
8	tons_surpluses	Numeric	The mass of edible food surplus generated. Surplus refers to edible food that is not consumed as intended (in tons).

9	tons_supply	Numeric	The total amount of food supplied that used to calculate surplus or waste rates.
10	us_dollars_surplus	Numeric	The economic value of the surplus food (in USD).
11	tons_waste	Numeric	The total amount of surplus food that became waste (in tons).
12	tons_uneaten	Numeric	The amount of surplus that was edible but not eaten.
13	tons_donations	Numeric	Quantity of surplus food redirected to donation programs or food banks.
14	tons_industrial_uses	Numeric	The amount of surplus food, in tons, that is redirected to industrial uses rather than being disposed of as waste.
15	tons_animal_feed	Numeric	Food surplus diverted to feeding livestock.
16	tons_anaerobic_digestion	Numeric	Surplus processed through anaerobic digestion to produce biogas.
17	tons_composting	Numeric	Surplus converted into compost through biological decomposition.
18	tons_not_harvested	Numeric	Food left unharvested at the farm stage due to quality issues, low prices, labor shortages or market rejection.
19	tons_incineration	Numeric	Surplus sent to facilities where it is burned for disposal or energy recovery.
20	tons_land_application	Numeric	Food discarded directly onto land.
21	tons_landfill	Numeric	Food waste that ends up in landfill sites.
22	tons_sewer	Numeric	Food waste disposed via sewer systems such as garbage disposals.
23	tons_dumping	Numeric	Food illegally dumped or discarded outside formal waste management systems.
24	surplus_upstream_100_year_mtco	Numeric	The 100-year upstream carbon footprint of surplus food. Includes emissions from

	2e_footprint		production, processing, and transport before consumer use (in metric tons of CO ₂ -equivalent).
25	surplus_downstream_100_year_mtco2e_footprint	Numeric	The 100-year downstream carbon footprint from disposal, waste handling, and decomposition after surplus occurs (in CO ₂ -equivalent).
26	surplus_total_100_year_mtco2e_footprint	Numeric	The combined upstream with downstream carbon footprint of surplus food over a 100-year period (in metric tons of CO ₂ -equivalent).
27	surplus_upstream_100_year_mtch4_footprint	Numeric	Upstream methane emissions over 100 years caused by surplus food production (in metric tons of CH ₄).
28	surplus_downstream_100_year_mtch4_footprint	Numeric	Downstream methane emissions (100-year timescale) generated by waste processing methods (in metric tons of CH ₄).
29	surplus_total_100_year_mtch4_footprint	Numeric	The total methane emissions generated from surplus food across all stages of the supply chain including both production and disposal processes (in metric tons of CH ₄).
30	gallons_water_footprint	Numeric	The total volume of water used to produce the surplus food (in gallons).
31	meals_wasted	Numeric	The number of meals equivalent lost due to food surplus becoming waste.

Table 2: Model Performance of XGBoost

Metric	Value
R ²	0.9996
Root Mean Squared Error (RMSE)	10.7377
Mean Absolute Error (MAE)	2.3750

Table 3: Top 10 Feature Importance of XGBoost

Feature	Importance
meals_wasted	0.54300
tons_uneaten	0.41777
sector_Retail	0.01662
sector_Manufacturing	0.00732

sector_Residential	0.00594
sector_Foodservice	0.00414
tons_surplus	0.00348
food_type_Ready-To-Drink Beverages	0.00038
state_Hawaii	0.00017
tons_supply	0.00016

Table 4: Model Performance of Random Forest

Metric	Value
R^2	0.9996
Root Mean Squared Error (RMSE)	10.62
Mean Absolute Error (MAE)	1.16

Table 5: Feature Importance of Random Forest

Feature	Importance
num_meals_wasted	0.514967
num_tons_uneaten	0.460366
cat_sector_Residential	0.008665
cat_sector_Retail	0.005369
cat_sector_Manufacturing	0.004988
num_tons_surplus	0.003748
num_surplus_total_100_year_mtch4_footprint	0.000441
num_surplus_total_100_year_mtco2e_footprint	0.000350
num_tons_supply	0.000331
cat_sector_Foodservice	0.000154

Table 6: Model Performance of FNN

Metric	Value
R^2	0.9777
Root Mean Squared Error (RMSE)	75.57
Mean Absolute Error (MAE)	25.02

Table 7: Model Performance of KiNN

Metric	Value
R^2	0.9324
Root Mean Squared Error (RMSE)	133.45
Mean Absolute Error (MAE)	46.38

Table 8: Comparison of Models

	XGBoost	Random Forest	FNN	KiNN
R^2	0.9996	0.9996	0.9777	0.9324
RMSE	10.7377	10.62	75.57	133.45
MAE	2.3750	1.16	25.02	46.38

APPENDIX B

Figure 1: Process Flowchart

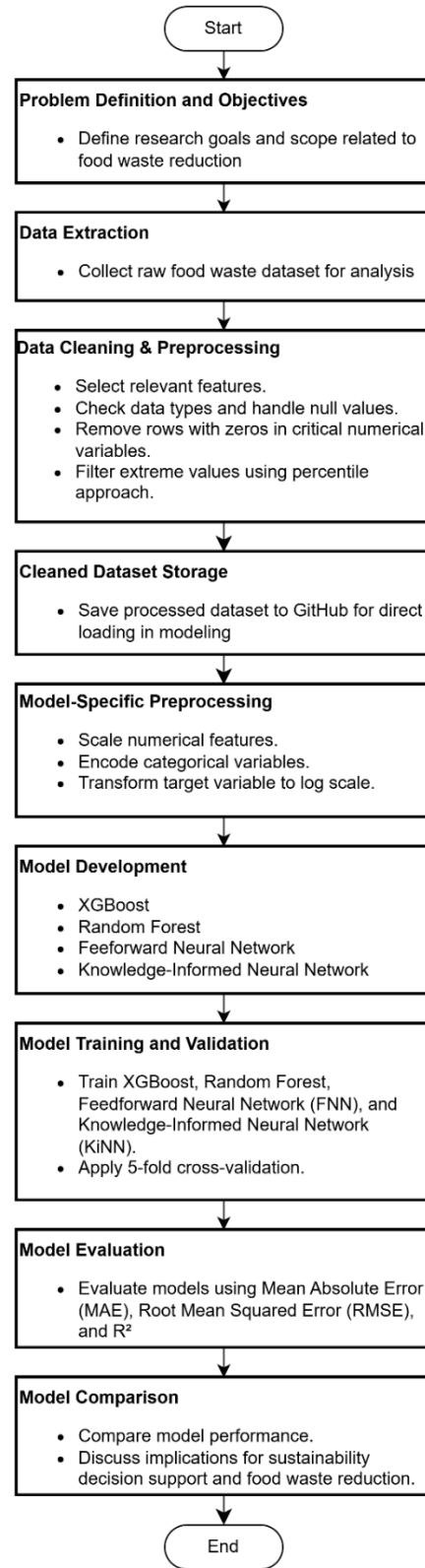


Figure 2: Top 10 Feature Importance in XGBoost Model

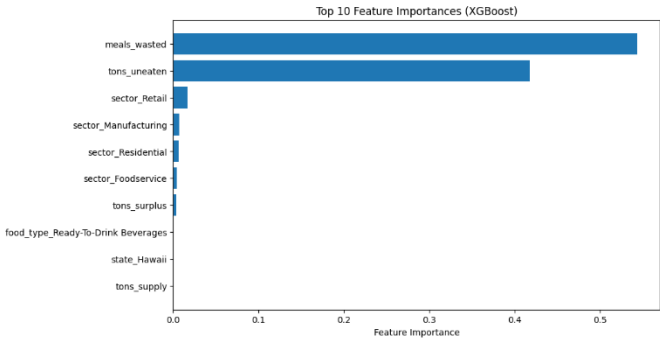


Figure 3: 5 Least Important Features in XGBoost Model

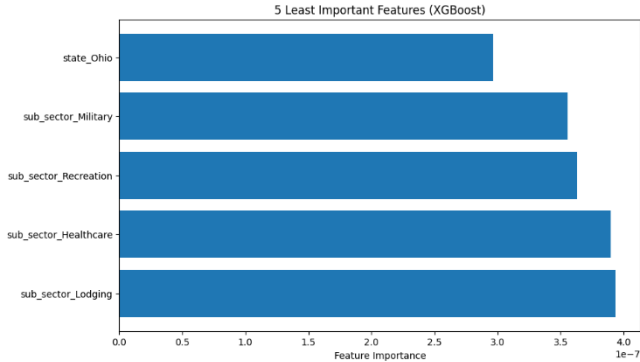


Figure 4: Top 10 Feature Importance in Random Forest Model

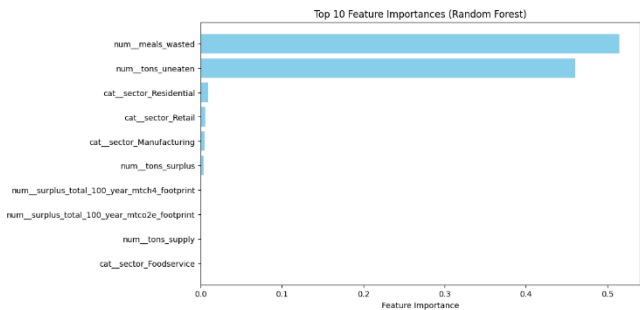
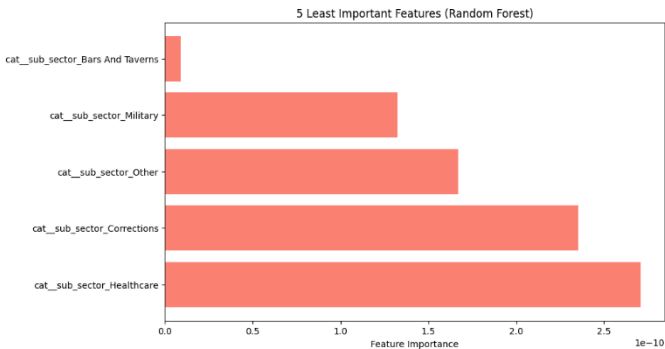


Figure 5: 5 Least Important Features in Random Forest Model



APPENDIX C

1. Raw ReFED Food Waste Dataset: <https://insights-engine.refed.org/food-waste-monitor?view=overview&year=2023>

2. Raw Dataset in Google Drive: <https://drive.google.com/file/d/12IX9yDyfyXAUoOXcmQLvRWwZrfX10KF/view?usp=sharing>
3. Cleaned Dataset and Codes on GitHub: https://github.com/Keertana0307/food_waste_cleaned_dataset